

Ruibin Yuan

PhD Student in Artificial Intelligence, HKUST | AI Researcher, Developer, and Musician

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Research focus. Foundation models for music generation and understanding; multimodal LLMs; large-scale data, training infrastructure, and open research communities for creative AI.

Education

Hong Kong University of Science and Technology (HKUST) <i>PhD in Artificial Intelligence</i> Advisor: Jie Fu, Wei Xue, Yike Guo	09/2023–Present Hong Kong
Carnegie Mellon University (CMU) <i>M.S. in Music and Technology, Computer Science Emphasis</i> Advisor: Roger B. Dannenberg, Gus Xia, Riccardo Schulz	09/2021–05/2023 Pittsburgh, PA
Beijing University of Posts and Telecommunications (BUPT) <i>B.S. in Electronic Science and Technology</i>	09/2016–06/2020 Beijing, China

Academic Experience

Qwen Music <i>Team Lead.</i> Lead the development of Qwen Music, a foundation model for music generation, and contribute to music understanding capabilities in the Qwen-Omni series.	04/2025–Present
Multimodal Art Projection Research Community (MAP) <i>Cofounder.</i> Build open large language, music, and multimodal models, with a focus on open-source training, data collection, and creative AI. Lead MAP’s multimodal and AI music research direction across projects including COIG, MERT, ChatMusician, CMMMU, and YuE.	09/2022–Present
YuE (OpenSuno) <i>Team Lead.</i> Built an open full-song generation foundation model designed to approach the capabilities of closed-source music generation systems such as Suno and Udio.	03/2024–03/2025
NEO LLM <i>Core Member.</i> Helped train a fully open LLM from scratch, contributing to infrastructure, PB-scale data collection and cleaning, training, evaluation, experiment logging, and tooling. Operated large GPU clusters and monitored hardware/software failures during training.	09/2023–05/2024
ChatMusician <i>Team Lead.</i> Trained the first symbolic music LLM; collected a large-scale symbolic music and language dataset and an advanced music understanding benchmark. The project received public attention from Yann LeCun and AK.	2023–2024
MMMU & CMMMU Contributed to multimodal benchmark development extending MMLU/CMMLU-style expert reasoning to multimodal settings. MMMU was recommended by Jeff Dean.	2023–2024
MERT Built a large-scale self-supervised BERT-style representation model for music audio.	2022–2023
MARBLE Built the first unified benchmark for music understanding.	2022–2023
COIG <i>Project Lead, COIG-PC.</i> Led the Chinese FLAN-style instruction tuning corpus effort, coordinating over one hundred contributors and converting thousands of pre-LLM-era Chinese NLP datasets into instruction format.	2022

Selected Publications and Manuscripts

Asterisks indicate equal contribution.

1. [CVPR 2024] MMMU: A Massive Multi-discipline Multimodal Understanding and Reasoning Benchmark for Expert AGI. Xiang Yue*, Yuansheng Ni*, Kai Zhang*, Tianyu Zheng*, Ruoqi Liu, Ge Zhang, Samuel Stevens, Dongfu Jiang, Weiming Ren, Yuxuan Sun, Cong Wei, Botao Yu, **Ruibin Yuan**, Renliang Sun, Ming Yin, Boyuan Zheng, Zhenzhu Yang, Yibo Liu, Wenhao Huang, Huan Sun, Yu Su, Wenhua Chen.

2. [ICLR 2024] MERT: Acoustic Music Understanding Model with Large-Scale Self-supervised Training. Yizhi Li*, **Ruibin Yuan***, Ge Zhang*, Yinghao Ma*, Xingran Chen, Hanzhi Yin, Chenghua Lin, Anton Ragni, Emmanouil Benetos, Norbert Gyenge, Roger Dannenberg, Ruibo Liu, Wenhui Chen, Gus Xia, Yemin Shi, Wenhao Huang, Yike Guo, Jie Fu.
3. [NeurIPS 2023] MARBLE: Music Audio Representation Benchmark for Universal Evaluation. **Ruibin Yuan***, Yinghao Ma*, Yizhi Li*, Ge Zhang*, Xingran Chen, Hanzhi Yin, Le Zhuo, Yiqi Liu, Jiawen Huang, Zeyue Tian, Binyue Deng, Ningzhi Wang, Chenghua Lin, Emmanouil Benetos, Anton Ragni, Norbert Gyenge, Roger Dannenberg, Wenhui Chen, Gus Xia, Wei Xue, Si Liu, Shi Wang, Ruibo Liu, Yike Guo, Jie Fu.
4. [ACL 2024 Main] AnyGPT: Unified Multimodal LLM with Discrete Sequence Modeling. Jun Zhan, Junqi Dai, Jiasheng Ye, Yunhua Zhou, Dong Zhang, Zhigeng Liu, Xin Zhang, **Ruibin Yuan**, Ge Zhang, Linyang Li, Hang Yan, Jie Fu, Tao Gui, Tianxiang Sun, Yugang Jiang, Xipeng Qiu.
5. [ACL 2024 Findings] ChatMusician: Understanding and Generating Music Intrinsically with LLM. **Ruibin Yuan**, Hanfeng Lin, Yi Wang, Zeyue Tian, Shangda Wu, Tianhao Shen, Ge Zhang, Yuhang Wu, Cong Liu, Ziya Zhou, Ziyang Ma, Liumeng Xue, Ziyu Wang, Qin Liu, Tianyu Zheng, Yizhi Li, Yinghao Ma, Yiming Liang, Xiaowei Chi, Ruibo Liu, Zili Wang, Pengfei Li, Jingcheng Wu, Chenghua Lin, Qifeng Liu, Tao Jiang, Wenhao Huang, Wenhui Chen, Emmanouil Benetos, Jie Fu, Gus Xia, Roger Dannenberg, Wei Xue, Shiyin Kang, Yike Guo.
6. [ISMIR 2023] On the Effectiveness of Speech Self-supervised Learning for Music. Yinghao Ma, **Ruibin Yuan**, Yizhi Li, Ge Zhang, Xingran Chen, Hanzhi Yin, Chenghua Lin, Emmanouil Benetos, Anton Ragni, Norbert Gyenge, Ruibo Liu, Gus Xia, Roger Dannenberg, Yike Guo, Jie Fu.
7. [ISMIR 2023] LyricWhiz: Robust Multilingual Zero-shot Lyrics Transcription by Whispering to ChatGPT. Le Zhuo, **Ruibin Yuan**, Jiahao Pan, Yinghao Ma, Yizhi Li, Ge Zhang, Si Liu, Roger B. Dannenberg, Jie Fu, Chenghua Lin, Emmanouil Benetos, Wenhui Chen, Wei Xue, Yike Guo.
8. [ISMIR 2022 LBD] MAP-Music2Vec: A Simple and Effective Baseline for Self-Supervised Music Audio Representation Learning. Yizhi Li*, **Ruibin Yuan***, Ge Zhang*, Yinghao Ma*, Chenghua Lin, Xingran Chen, Anton Ragni, Hanzhi Yin, Zhijie Hu, Haoyu He, Emmanouil Benetos, Norbert Gyenge, Ruibo Liu, Jie Fu.
9. [Manuscript] ComposerX: Multi-Agent Symbolic Music Composition with LLMs. Qixin Deng, Qikai Yang, **Ruibin Yuan**, Yipeng Huang, Yi Wang, Xubo Liu, Zeyue Tian, Jiahao Pan, Ge Zhang, Hanfeng Lin, Yizhi Li, Yinghao Ma, Jie Fu, Chenghua Lin, Emmanouil Benetos, Wenwu Wang, Guangyu Xia, Wei Xue, Yike Guo.
10. [CVPR 2024] Weakly-Supervised Emotion Transition Learning for Diverse 3D Co-speech Gesture Generation. Xingqun Qi, Jiahao Pan, Peng Li, **Ruibin Yuan**, Xiaowei Chi, Mengfei Li, Wenhan Luo, Wei Xue, Shanghang Zhang, Qifeng Liu, Yike Guo.
11. [DMRN+17] MAP-Mu-BERT: Large-Scale Pretrained Model for Self-Supervised Music Audio Representation Learning. Yizhi Li*, **Ruibin Yuan***, Ge Zhang*, Yinghao Ma*, Chenghua Lin, Xingran Chen, Anton Ragni, Hanzhi Yin, Zhijie Hu, Haoyu He, Emmanouil Benetos, Norbert Gyenge, Ruibo Liu, Jie Fu.
12. [ACL 2024 Findings] CIF-Bench: A Chinese Instruction-Following Benchmark for Evaluating the Generalizability of Large Language Models. Yizhi Li, Ge Zhang, Xingwei Qu, Jiali Li, Zhaoqun Li, Zekun Wang, Hao Li, **Ruibin Yuan**, Yinghao Ma, Kai Zhang, Wangchunshu Zhou, Yiming Liang, Lei Zhang, Lei Ma, Jiajun Zhang, Zuowen Li, Stephen W. Huang, Chenghua Lin, Wenhui Chen, Jie Fu.
13. [Interspeech 2022] DeID-VC: Speaker De-identification via Zero-shot Pseudo Voice Conversion. **Ruibin Yuan**, Yuxuan Wu, Jacob Li, Jaxter Kim.
14. [Manuscript] CMMU: A Chinese Massive Multi-discipline Multimodal Understanding Benchmark. Ge Zhang, Xinrun Du, Bei Chen, Yiming Liang, Tongxu Luo, Tianyu Zheng, Kang Zhu, Yuyang Cheng, Chunpu Xu, Shuyue Guo, Haoran Zhang, Xingwei Qu, Junjie Wang, **Ruibin Yuan**, Yizhi Li, Zekun Wang, Yudong Liu, Yu-Hsuan Tsai, Fengji Zhang, Chenghua Lin, Wenhao Huang, Wenhui Chen, Jie Fu.
15. [Manuscript] MuPT: A Generative Symbolic Music Pretrained Transformer. Xingwei Qu, Yuelin Bai, Yinghao Ma, Ziya Zhou, Ka Man Lo, Jiaheng Liu, **Ruibin Yuan**, Lejun Min, Xueling Liu, Tianyu Zhang, Xinrun Du, Shuyue Guo, Yiming Liang, Yizhi Li, Shangda Wu, Juntong Zhou, Tianyu Zheng, Ziyang Ma, Fengze Han, Wei Xue, Gus Xia, Emmanouil Benetos, Xiang Yue, Chenghua Lin, Xu Tan, Stephen W. Huang, Wenhui Chen, Jie Fu, Ge Zhang.
16. [Manuscript] COIG-CQIA: Quality is All You Need for Chinese Instruction Fine-tuning. Yuelin Bai, Xinrun Du, Yiming Liang, Yonggang Jin, Ziqiang Liu, Juntong Zhou, Tianyu Zheng, Xincheng Zhang, Nuo Ma, Zekun Wang, **Ruibin Yuan**, Haihong Wu, Hongquan Lin, Wenhao Huang, Jiajun Zhang, Wenhui Chen, Chenghua Lin, Jie Fu, Min Yang, Shiwen Ni, Ge Zhang.
17. [Manuscript] The Fine Line: Navigating Large Language Model Pretraining with Downstream Capability Analysis. Chen Yang, Junzhuo Li, Xinyao Niu, Xinrun Du, Songyang Gao, Haoran Zhang, Zhaoliang Chen, Xingwei Qu, **Ruibin Yuan**, Yizhi Li, Jiaheng Liu, Stephen W. Huang, Shawn Yue, Wenhui Chen, Jie Fu, Ge Zhang.

Industry Experience

Qwen	04/2025–Present
Research Intern; Supervisor: Jin Xu	Remote
Moonshot.ai	09/2024–04/2025
Research Intern; Supervisor: Xu Tan	Remote
BAAI	2022–2023
Part-time Research Intern; Supervisor: Jie Fu	Remote

Stardust.ai <i>Part-time ML Consultant; Ex-engineer</i>	05/2020–2021 Beijing, China
NetEase, Inc. <i>Intern Machine Learning Engineer</i>	12/2019–04/2020 Shanghai, China
Vlahakis Recording Studio at CMU <i>Student Audio Engineer</i>	2023 Pittsburgh, PA

Music Experience

Lead Guitarist and Cofounder, BUPT Slow Blues Band [Album]	2016–2021
President, BUPT Guitar Club	2017–2018